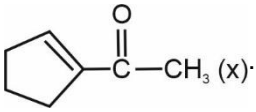
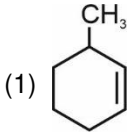
CHEMISTRY

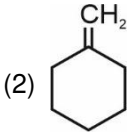
SECTION - A

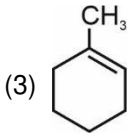
Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

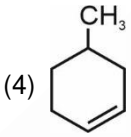
Choose the correct answer :

51. Aman has been asked to synthesise the molecule  (x). He thought of preparing the molecule using an aldol condensation reaction. He found a few cyclic alkenes in his laboratory. He thought of performing ozonolysis reaction on alkene to produce a dicarbonyl compound followed by aldol reaction to prepare "x". Predict the suitable alkene that can lead to the formation of "x".

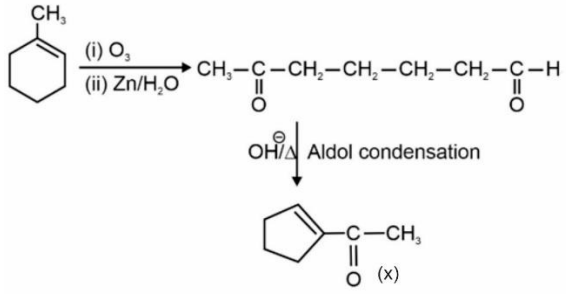
(1) 

(2) 

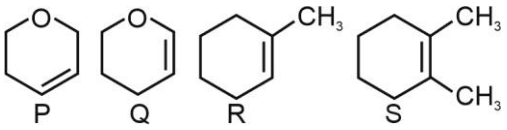
(3) 

(4) 

Answer (3)

Sol. 

52. Following are the four molecules "P", "Q", "R" and "S". Which one among the four molecules will react with H–Br(aq) at the fastest rate?

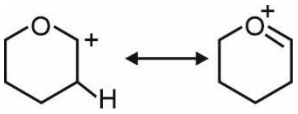


(1) P (2) R

(3) Q (4) S

Answer (3)

Sol. The reaction with H — Br will be fastest for the specie forming most stable carbocation



Due to resonance, it will have most stable carbocation.

53. One mole of the octahedral complex compound $Co(NH_3)_5Cl_3$ gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with excess of $AgNO_3$ solution to yield two moles of AgCl(s). The structure of the complex is

(1) $[Co(NH_3)_4Cl].Cl_2.NH_3$

(2) $[Co(NH_3)_4Cl_2].Cl.NH_3$

(3) $[Co(NH_3)_3Cl_3].2NH_3$

(4) $[Co(NH_3)_5Cl]Cl_2$

Answer (4)

Sol. $[Co(NH_3)_5Cl]Cl_2 \xrightarrow{H_2O} \underbrace{[Co(NH_3)_5Cl]^{2+} + 2Cl^-}_{3 \text{ ions}}$

$\downarrow + AgNO_3$

2 mole AgCl

54. The carbohydrate "Ribose" present in DNA, is

A. A pentose sugar

B. present in pyranose form

C. in "D" configuration

D. a reducing sugar, when free

E. in α – anomeric form

Choose the correct answer from the options given below

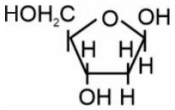
(1) A, B and E only (2) B, D and E only

(3) A, C and D only (4) A, D and E only

Answer (3)



Sol. Ribose present in DNA is β-(D)-deoxyribose



It is a pentose sugar present in furanose form having D configuration.

It is a reducing sugar having β-anomeric form

A, C & D are correct

55. Which of the following statement is true with respect to H_2O, NH_3 and CH_4?

A. The central atoms of all the molecules are sp^3 hybridized.

B. The H – O – H, H – N – H and H – C – H angles in the above molecules are 104.5º, 107.5º and 109.5º, respectively.

C. The increasing order of dipole moment is $CH_4 < NH_3 < H_2O$.

D. Both H_2O and NH_3 are Lewis acids and CH_4 is a Lewis base.

E. A solution of NH_3 in H_2O is basic. In this solution NH_3 and H_2O act as Lowry-Bronsted acid and base respectively

Choose the correct answer from the options given below

(1) A, B and C only (2) A, B, C and E only

(3) C, D and E only (4) A, D and E only

Answer (1)

Sol.

	H_2O	NH_3	CH_4
Hybridization	sp^3	sp^3	sp^3
Bond angle	104.5º	107.5º	109.5º
Dipole moment	1.85	1.47	0

Both H_2O & NH_3 are Lewis bases. NH_3 acts as base in H_2O, NH_3 & H_2O act as Lowry – Bronsted base & acid respectively.

A, B & C are correct

56. Which of the following statements are NOT true about the periodic table?

A. The properties of elements are function of atomic weights.

B. The properties of elements are function of atomic numbers.

C. Elements having similar outer electronic configuration are arranged in same period.

D. An element's location reflects the quantum numbers of the last filled orbital.

E. The number of elements in a period is same as the number of atomic orbitals available in energy level that is being filled.

Choose the correct answer from the options given below.

(1) D and E only (2) B, C and E only

(3) A and E only (4) A, C and E only

Answer (4)

Sol. Properties of elements are function of atomic numbers.

Elements having similar outer electronic configuration are arranged in same group.

Number of elements in a period is double of number of atomic orbitals available in energy level that is being filled.

So, (A), (C) and (E) are incorrect.

57. The large difference between the melting and boiling points of oxygen and sulphur may be explained on the basis of

(1) Atomicity

(2) Electronegativity

(3) Atomic size

(4) Electron gain enthalpy

Answer (1)

Sol. The large difference in the melting and boiling points of oxygen and sulphur is due to atomicity as oxygen exists as O_2 and sulphur exists as S_8.

58. For a reaction, $N_2O_5(g) \rightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$ in a constant volume container, no products were present initially. The final pressure of the system when 50% of reaction gets completed is

(1) 5 times of initial pressure

(2) 5/2 times of initial pressure

(3) 7/4 times of initial pressure

(4) 7/2 times of initial pressure

Answer (3)



Sol.

$$N_2O_5(g) \rightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$$

$t = 0$	P_0		
$t = t_{50\%}$	$P_0(1-\alpha)$	$2P_0\alpha$	$\frac{P_0\alpha}{2}$

$$\text{Final pressure} = P_0 + P_0\alpha + \frac{P_0\alpha}{2} \quad (\alpha = 0.5)$$

$$= P_0 + \frac{P_0}{2} + \frac{P_0}{4}$$

$$= P_0\left(1 + \frac{1}{2} + \frac{1}{4}\right)$$

$$= \frac{7P_0}{4}$$

59. Given below are two statements I and II.

Statement I : Dumas method is used for estimation of "Nitrogen" in an organic compound.

Statement II : Dumas method involves the formation of ammonium sulphate by heating the organic compound with conc. H_2SO_4.

In the light of the above statements, choose the ***correct*** answer from the options given below.

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

Answer (3)

Sol. Dumas method is used to estimate nitrogen which involves heating of compound with copper oxide.

60. Let us consider an endothermic reaction which is non-spontaneous at the freezing point of water. However, the reaction is spontaneous at boiling point of water. Choose the correct option.

(1) Both ΔH and ΔS are (–ve)

(2) Both ΔH and ΔS are (+ve)

(3) ΔH is (+ve) but ΔS is (–ve)

(4) ΔH is (–ve) but ΔS is (+ve)

Answer (2)

Sol. On increasing temperature, the given endothermic reaction becomes spontaneous.

$\Delta G = \Delta H - T\Delta S$

For spontaneity,

$\Delta G < 0$

Given,

$\Delta H > 0$

So, ΔS should be +ve.

61. Which of the following linear combination of atomic orbitals will lead to formation of molecular orbitals in homonuclear diatomic molecules [internuclear axis in z-direction] ?

A. $2p_z$ and $2p_x$

B. 2s and $2p_x$

C. $3d_{xy}$ and $3d_{x^2-y^2}$

D. 2s and $2p_z$

E. $2p_z$ and $3d_{x^2-y^2}$

Choose the correct answer from the options given below:

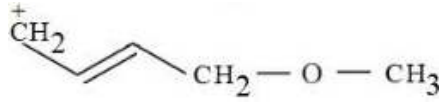
(1) A and B Only (2) D Only

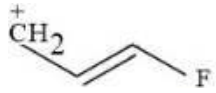
(3) E Only (4) C and D Only

Answer (2)

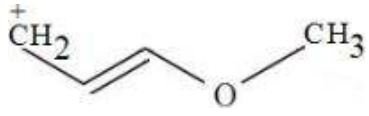
Sol. 2s and $2p_z$ will lead to formation of molecular orbital as they have very small difference in energies.

62. Which one of the carbocations from the following is most stable?

(1) 

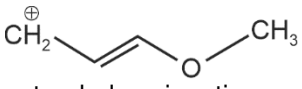
(2) 

(3) 

(4) 

Answer (4)



Sol.  will be most stable due to extended conjugation

(3) will be less stable due to –M of 

63. Given below are two statements:

Statement I: The conversion proceeds well in the less polar medium.

$$CH_3 - CH_2 - CH_2 - CH_2 - Cl \xrightarrow{HO^-} CH_3 - CH_2 - CH_2 - CH_2 - OH + Cl^{(-)}$$

Statement II: The conversion proceeds well in the more polar medium.

$$CH_3 - CH_2 - CH_2 - CH_2 - Cl \xrightarrow{R_3\ddot{N}} CH_3 - CH_2 - CH_2 - CH_2 - \overset{\overset{R}{|}}{\underset{\underset{R}{|}}{N^{(+)}}} - R\ Cl^{(-)}$$

In the light of the above statements, choose the ***correct*** answer from the options given below

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

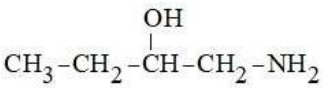
(4) Both Statement I and Statement II are false

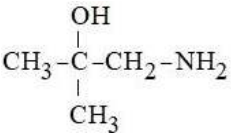
Answer (3)

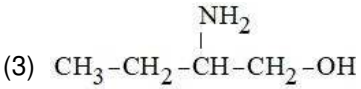
Sol. Both the reactions will proceed via SN^2 mechanism but OH^- will proceed in less polar medium while $R_3\ddot{N}$ will occur well in more polar medium. Both statements are true.

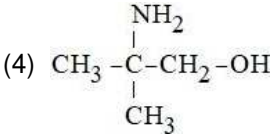
64. The product (A) formed in the following reaction sequence is

$$CH_3 - C \equiv CH \xrightarrow[\text{iii) } H_2/Ni]{\text{i) } Hg^{2+}, H_2SO_4 \text{ ii) HCN}} \underset{\text{Product}}{(A)}$$

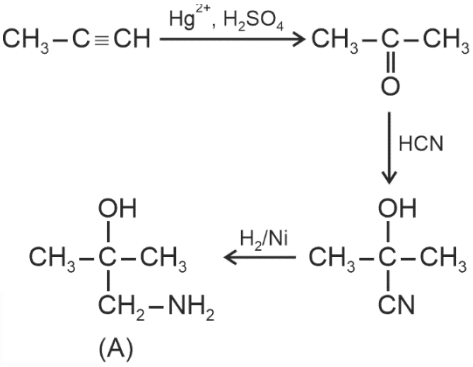
(1) 

(2) 

(3) 

(4) 

Answer (2)

Sol. 

(A)

65. K_{sp} for $Cr(OH)_3$ is 1.6×10^{-30}. What is the molar solubility of this salt in water?

(1) $\sqrt[5]{1.8\times10^{-30}}$ (2) $\sqrt[4]{\frac{1.6\times10^{-30}}{27}}$

(3) $\frac{1.8\times10^{-30}}{27}$ (4) $\sqrt[2]{1.6\times10^{-30}}$

Answer (2)

Sol. $Cr(OH)_3 \rightleftharpoons \underset{s}{Cr^{3+}}(aq) + \underset{3s}{3OH^-}(aq)$

$K_{sp} = s(3s)^3$

$1.6 \times 10^{-30} = 27s^4$

$\sqrt[4]{\frac{1.6\times10^{-30}}{27}} = s$

66. Which of the following ions is the strongest oxidizing agent?

(Atomic Number of Ce = 58, Eu = 63, Tb = 65, Lu = 71)

(1) Ce^{3+} (2) Eu^{2+}

(3) Tb^{4+} (4) Lu^{3+}

Answer (3)

Sol. Tb^{4+} is strongest oxidising agent as it will reduce to Tb^{3+} (common O.S. of Ln)



67. Which of the following arrangements with respect to their reactivity in nucleophilic addition reaction is correct?

(1) p-nitrobenzaldehyde < benzaldehyde < p-tolualdehyde < acetophenone

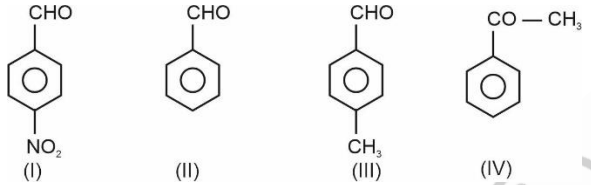
(2) acetophenone < p-tolualdehyde < benzaldehyde < p-nitrobenzaldehyde

(3) benzaldehyde < acetophenone < p-nitrobenzaldehyde < p-tolualdehyde

(4) acetophenone < benzaldehyde < p-tolualdehyde < p-nitrobenzaldehyde

Answer (2)

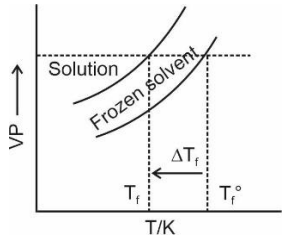
Sol. Reactivity $\propto$ +ve charge on electrophilic carbon



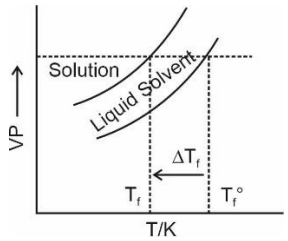
Correct order I > II >III > IV

68. Consider the given plots of vapour pressure (VP) vs temperature (T/K). Which amongst the following options is correct graphical representation showing ΔT_f. depression in the freezing point of a solvent in as solution?

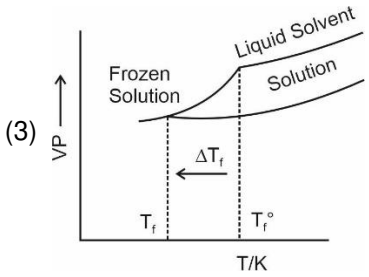
(1)



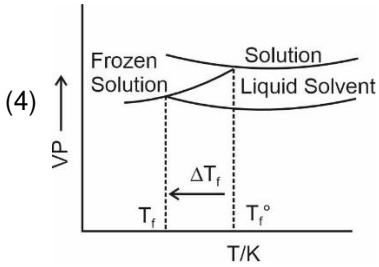
(2)



(3)

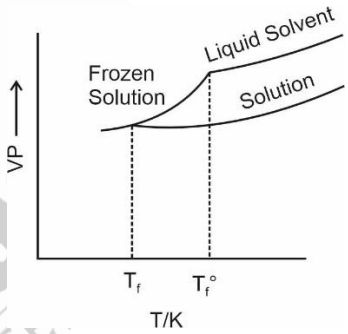


(4)



Answer (3)

Sol. Correct variation



69. For the given cell

$Fe^{2+}(aq) + Ag^{+}(aq) \rightarrow Fe^{3+}(aq) + Ag(s)$

The standard cell potential of the above reaction is

Given:

$Ag^{+} + e^{-} \rightarrow Ag$ $\quad E^{\theta} = x\ V$

$Fe^{2+} + 2e^{-} \rightarrow Fe$ $\quad E^{\theta} = y\ V$

$Fe^{3+} + 3e^{-} \rightarrow Fe$ $\quad E^{\theta} = z\ V$

(1) x + y – z (2) x + 2y

(3) x + 2y – 3z (4) y – 2x

Answer (3)



Sol.

$Ag^+ + e^- \longrightarrow Ag$ $\quad E^\circ = x\ V$ $\quad \Delta G^\circ_1 = -1 \times (x) \times F$

$Fe^{2+} + 2e^- \longrightarrow Fe$ $\quad E^\circ = y\ V$ $\quad \Delta G^\circ_2 = -2 \times (y) \times F$

$Fe \longrightarrow Fe^{3+} + 3e^-$ $\quad E^\circ = z\ V$ $\quad \Delta G^\circ_3 = -2 \times (-z) \times F$

$Ag^+ + Fe^{2+} \longrightarrow Fe^{3+} + Ag$

$\Delta G^\circ = \Delta G^\circ_1 + \Delta G^\circ_2 + \Delta G^\circ_3$

$-1 \times E^\circ \times F = -1\ (x)\ F - 2(y)\ F + 3(z)F$

$E^\circ = x + 2y - 3z$

70. Preparation of potassium permanganate from MnO_2 involves two step process in which the 1st step is a reaction with KOH and KNO_3 to produce

(1) K_2MnO_4 (2) $K_4(Mn(OH)_6]$

(3) $KMnO_4$ (4) K_3MnO_4

Answer (1)

Sol. Preparation of $KMnO_4$

$$MnO_2 \xrightarrow[KNO_3]{KOH} K_2MnO_4 \xrightarrow[\text{Oxidation in Alkaline solution}]{\text{Electrolytic}} KMnO_4$$

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

71. Among the following cations, the number of cations which will give characteristic precipitate in their identification tests with $K_4[Fe(CN)_6]$ is ____.

$Cu^{2+}, Fe^{3+}, Ba^{2+}, Ca^{2+}, NH_4^+, Mg^{2+}, Zn^{2+}$

Answer (3)

Sol. Cu^{2+}, Ca^{2+} & Zn^{2+} will form precipitate with $K_4[Fe(CN)_6]$. Fe^{3+} forms coloured solution

72. 37.8 g N_2O_5 was taken in a 1 L reaction vessel and allowed to undergo the following reaction at 500 K

$$2N_2O_5(g) \rightleftharpoons 2N_2O_4(g) + O_2(g)$$

The total pressure at equilibrium was found to be 18.65 bar.

Then, Kp = _____ $\times 10^{-2}$ [nearest integer]

Assume N_2O_5 to behave ideally under these conditions.

Given : R = 0.082 bar L mol^{-1} K^{-1}

Answer (962)

Sol. $2N_2O_5(g) \rightleftharpoons 2NO_2(g) + O_2(g)$

$$P_i = \frac{n_i R \times T}{V_i}$$

$$= \frac{\frac{37.8}{108} \times 0.082 \times 500}{1}$$

= 14.35 atm

	$2N_2O_5(g)$ $\rightleftharpoons$	$2NO_2$ +	$O_2(g)$
$t = 0$	14.35		
$t = t_{eq}$	$14.35 - 2P$	$2P$	P

$P_f = 14.35 + P = 18.65$

$P = 4.3$ atm

$$Kp = \frac{(8.6)^2 \times (4.3)}{(5.75)^2}$$

= 9.62 bar

= 962 $\times 10^{-2}$ bar

73. Standard entropies of X_2, Y_2 and XY_5 are 70, 50 and 110 J K^{-1} mol^{-1} respectively. The temperature in Kelvin at which the reaction

$$\frac{1}{2}X_2 + \frac{5}{2}Y_2 \rightleftharpoons XY_5 \quad \Delta H^\circ = -35 kJ mol^{-1}$$

will be at equilibrium is _____. (Nearest integer)

Answer (700)



Sol. $\frac{1}{2}X_2 + \frac{5}{2}Y_2 \rightleftharpoons XY_5$

For equilibrium

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$0 = -35\times10^3 - T\left(110 - \left(\frac{5}{2}\times50\right) - \frac{1}{2}\times70\right)$

$35 \times 10^3 = 50 \times T$

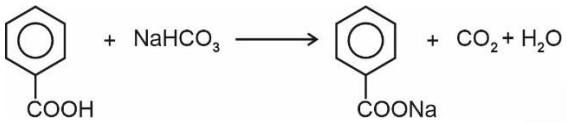
$700K = T$

74. X g of benzoic acid on reaction with aq. $NaHCO_3$ released CO_2 that occupied 11.2 L volume at STP.

X is _____ g.

Answer (61)

Sol.



1 mole benzoic acid will release 1 mole CO_2

11.2 L CO_2 at STP is $\frac{1}{2}$ mole CO_2 which will be released by reaction with $\frac{1}{2}$ mole benzoic acid

Mass of benzoic acid = $\frac{1}{2}\times122$

= 61 gm

75. Consider the following reaction occurring in the blast furnace :

$Fe_3O_4(s) + 4CO(g) \rightarrow 3Fe(l) + 4CO_2(g)$

'x' kg of iron is produced when 2.32×10^3 kg Fe_3O_4 and 2.8×10^2 kg CO are brought together in the furnace. The value of 'x' is _____. (Nearest integer)

[Given :

molar mass of Fe_3O_4 = 232 g mol^{-1}

molar mass of CO = 28 g mol^{-1}

molar mass of Fe = 56 g mol^{-1}]

Answer (420)

Sol.

	$Fe_3O_4(s)$	+	$4CO(g)$	→	$3Fe(l) + 4CO_2(g)$
moles taken	$\frac{2.32\times10^6}{232}$		$\frac{2.8\times10^5}{28}$		

CO is Limiting reagent

Iron produced (in kg) = $\frac{3}{4}\times\frac{2.8}{28}\times10^5\times56\times10^{-3}$

= 420 kg

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